Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	**Name**	*Egbers, Kees*	**Email**	*Kees.egbers@ipt.fraunhofer.de*
Institute Name	*Fraunhofer Institute for Production Technology IPT*			
Address	*Steinbachstrasse 17, 52074 Aachen, Germany*			

2. Technology Overview

Title of Proposed Technology	*Development of an enhanced sheet metal forming process with (ultra-)sonic-vibrations*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery (), Advanced Mobility (), Next-gen Nuclear Power (), Advanced Bio (), Aerospace/Marine (), Hydrogen (√), Cybersecurity (), AI (), Next-gen Communication (), Advanced Robotics & Manufacturing (√), Quantum ()			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical and materials		*Mechanical and materials*	
	Medium Category	Small Category	Medium Category	Small Category
	Plastic Deformation/ Powder	*Sheet Forming Technologies*	*Precision manufacturing machinery*	*Press Machines*
Concept of Proposed Technology	*Increasing the formability of sheet metals by inducing (ultra-) sonic vibrations into the forming tools. Additional benefits of this technology include better surface quality of the formed part and lower forming forces compared to forming without vibration.*			
Rationale for Joint R&D Project	*Ultrasonic forming is proven as a concept for force reduction and improved formability, but forming of larger, commercially viable samples has still to be demonstrated (1).* *The same working principle applies to sheet metal bending, which has yet to be tested with ultrasonic assistance and commercially sized samples. The development of new actuators is needed to form commercially viable sample sizes. Therefore, cooperation with industrial partners is essential for successful development and validation. The technology is aimed at specific components during a fist phase — namely, metallic bipolar plates for fuel cells (FC) and electrolyzers (EL) and battery electrodes. There are different concepts for those components on the German and Asian markets, particularly in Korea. These differences affect key properties such as component size, material and thickness, geometry, the process chain, and operational and production standards. The collaboration aims to ensure the technical and regulatory applicability of the developed technology, before breaching out to other precision formed components.* *Cooperation is also essential from a technical perspective for implementing and validating the content of the research*			

Kommentiert [MA1]: Kooperation with industrial partners

	project. Successful implementation requires expertise in precise thin sheet forming processes, product manufacturing and market requirements, mechanical design (storage, assembly and guidance of components), sensor technology (for measuring process variable during the sonic-assisted forming process) and control technology (for compensating for deviations in materials and process variables). Such expertise can only be fully represented by a research consortium.
Key Words	*1. Sonic forming & ultrasonic forming of thin metal sheets and bipolar plates for fuel cells and electrolyzers* *2. Innovative forming process for sheet metal* *3. Increasing formability in cold forming processes* *4. Force reduction during forming processes*

Kommentiert [MA2]: + thin metal sheets

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6)

R&D Objectives

1. Simulation model tailored towards material flow under ultrasonic vibration
2.Concept of scalable ultrasonic actuator-setups for stamping& embossing of sheet metal
3. Concept of ultrasonic actuators for press brakes
4. Development of ultrasonic actuator-setups for stamping & embossing
5. Development of ultrasonic actuators for press brakes
6. Buildup and validation of prototypic ultrasonic actuators

R&D Target Specifications

Organizational:
1. Developed module for sonic-assisted forming is ready to transfer to Korean partner for production of sheet metal components
Technical
1. Reduction of required press tonnage for bipolar plate forming by 20%
- Enabling existing machinery to form Fuel Cell & Electrolyzer Bipolar Plates
2. Improving surface quality by reducing surface roughness of formed components by 30%
3. Improving forming limits of thin metals sheets by 5%

R&D Plan

Specific Works	1st Year				2nd Year				3rd Year				Final Year	
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2
1. Feasibility study														
1-1. Research of feasible process parameter sets														
1-2. Simulation of material flow under vibration														
1-3. Pilot tests														
2. Stamping & Embossing setup														
2-1. Concept														
2-2. Development & design of scalable actuator														
2-3. Lab-Scale actuator element validation														
2-4. Sensor integration														
2-5. Build-up of scaled prototype														
2-6. Validation tests														
3. Bending setup														
3-1. Concept														
3-2. Development & design of scalable actuator														
3-3. Lab-Scale actuator element validation														
3-4. Sensor integration														
3-3. Build-up of prototype														
3-4. Validation tests														

The technology is to be developed through the following steps:
1. Define the requirements for the product, e.g. component size, material thickness and component geometry, as well as production-related key figures such as cycle time and process-critical manufacturing deviations.
2. Pilot test and definition of suitable process parameter sets for the selected product, like a specific fuel cell bipolar plate
3. Selection, development and integration of suitable actuator elements for sonic-assisted forming.
Milestone 1: Completion of the mechanical construction of a scalable actuator element.
4. Validation of the actuator elements at lab scale
5. Integration of measurement technology to detect process deviations and forming forces, particularly uneven distribution of pressure force or misalignment between stamp and die
Milestone 2: Measurement technology for detecting process deviations integrated.
5. Integration of multiple actuator element into a commercially sized actuator
6. Commissioning of the device and development/parameterization of the control loop to compensate process

	deviations. *Milestone 3: Active compensation of process deviations enabled.* *7. Validation of the technology at an industrially relevant scale.*

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*To implement the project within the consortium, an equal and cooperative distribution is planned. A precise division of work packages and roles should be made based on the specific competencies within the consortium.* *Fraunhofer IPT has extensive experience of the entire sheet metal processing chain, including forming, laser processing, toolmaking and quality control. This includes mechanical development, control engineering, commissioning and process development. In particular, the Fraunhofer Institute contributes its expertise to the mechanical development of the device and the forming process. It will lead the build-up of the test bench and the large-scale actuators, including commissioning and validation. The press systems at Fraunhofer IPT with up to 800 tons of forming force can be used to conduct the commercial scale validation.*
Expected R&R of Korean SME Partner	*There are particular opportunities in developing and selecting suitable components of the actuator elements and control system. Additionally, the various market access requirements and product specifications in European and Asian markets, particularly Korea, must be met to ensure the technology's universal application.* *To ensure the easy adoption of the developed technology, the SME partner will provide a requirement list based on the existing forming equipment and forming operations of the company or their customers. The partner will also oversee the development and selection of the actuator components to ensure compatibility with Korean supply chains. The validation of the final prototype can be conducted at the facility of the partner.* *A precise division of work packages and roles should be made based on the specific competencies within the consortium.*

5. Expected Outcomes/Impact

Expected Outcomes and Impact	*Technological achievements:* *- Simulation model for material flow during forming with acoustic softening effects* *- Technology proven for commercial application by build-up and validation of scaleable demonstrator modules for BPP forming* *- Reduction of forming force for stainless steel fuel cell bipolar plates by 20%* *- Improved surface quality due to lower roughness of formed part* *- Build-up and validation of press brake demonstrator with build in vibration modules* *- Reduction of minimal bending radius for aluminum sheet metal of 20%* *Commercialization potential:* *- Commercialization of actuator modules for retrofitting presses and press brakes for components with high press forces* *- Development and commercialization of high performance precision forming stations* *Economic impact:* *- Enable local industries to compete in sectors, where precision formed and or lightweight components are needed, such as automotive, hydrogen and battery production.*

6. References

(1) *(Shao, G., Li, H., Zhan, M. (2021). A Review on Ultrasonic-Assisted Forming: Mechanism, Model, and Process. Chinese Journal of Mechanical Engineering. https://doi.org/10.1186/s10033-021-00612-0)*

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization